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Boron additive passivated carbonate electrolytes for stable cycling of 5 V lithium–metal batteries†

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The limitation of a high-voltage lithium (Li) metal battery lies in the absence of a robust electrolyte that can endure oxidation loss at a high-voltage cathode and suppress the dendrite growth at a Li metal anode. The challenge becomes especially serious when the cathode voltage reaches 5 V and above. Here, we developed an additive-containing electrolyte that can concurrently stabilize the Li metal anode and the 5 V class $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$ (LNMO) cathode by introducing tris(pentafluorophenyl)borane (TPFPB) to the traditional ethylene carbonate/ethyl methyl carbonate (EC/EMC) electrolyte. The boron element and pentafluorobenzene in TPFPB induce preferential formation of a steady and smooth interfacial layer on both the Li and LNMO electrodes, subsequently preventing the decomposition of the electrolyte, impeding the pulverization of the cathode electrode, and retarding the growth of Li dendrites. Accordingly, we accomplish high average coulombic efficiency of >99.2% and superior capacity retention of 90% after 500 cycles at 0.5C with charge cut-off voltage of 4.95 V, which is one of the highest results for cycling among 5 V-class Li metal batteries with 90% capacity retention. This study provides essential technological implications for the electrolyte design for high-voltage Li–metal batteries.

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Introduction

Lithium-ion (Li-ion) batteries have dominated the energy storage market in the past decades, but the increasing demands for energy density (e.g., for power electrical vehicles) call for a new energy storage technology.^{1,2} This circumstance revokes the prosperity of Li–metal batteries (LMBs) given the ultrahigh theoretical specific capacity of Li metal (3860 mA h g^{-1}), which is an order of magnitude larger than that of the commercialized graphite anode (372 mA h g^{-1}). To approach the maximum energy density, Li metal anode needs to be matched with a high-voltage cathode, for instance, a 5 V $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$ (LNMO) cathode, which has a high working potential, excellent rate performance, outstanding gravimetric energy density, and low cost. One significant challenge of high-voltage LMBs is the lack of a proper electrolyte that can offer predesigned interfacial

chemistry to stabilize both the Li anode and the high-voltage cathode while simultaneously sustaining the high voltage environment. This dilemma has encouraged pervasive explorations for new electrolytes with high working potential and satisfactory electrode protecting performance. For certain cathode materials, unconventional electrolytes including sulfone-based solvents, ionic liquids, and dinitrile solvents were found effective in the resistance of oxidation loss.^{3–6} However, these electrolytes are expensive and usually encounter difficulties while matching with the Li metal anode. For the Li–metal end, concentrated salt electrolytes and solid electrolytes show certain effectiveness in suppressing the growth of Li dendrites,^{7,8} but their low conductivity and high cost exclude them from a scale up approach.

Aside from pursuing new systems, a more effective and economical option is to introduce electrolyte additives to conventional ether or carbonate electrolytes.^{9–11} The contemporary aim of these additives is to help the formation of an electrode–electrolyte interface that protects the conventional electrolyte from decomposition and prevents considerable deterioration of electrode materials. Commonly used additives include tris(trimethylsilyl)phosphate (TMSP),¹² 1,10-sulfonyldiimidazole (SDM),¹³ tris(trimethylsilyl)phosphite (TMSPi),¹⁴ vinylene carbonate (VC),¹⁵ and phenyl vinyl sulfone (PVS).¹⁶ These additives are only beneficial to one electrode and cannot be synchronously reacted to form protective thin films on both electrodes. Recently, Xu *et al.* reported a “duo-effect” additive that can produce joint passivations on both electrodes for an

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ether electrolyte system with a charge cut-off voltage of 4.3 V.¹⁷ The relatively low intrinsic decomposition potential of the ether electrolyte (4 V) nevertheless constrains further improvement in the operational voltage,¹⁸ leaving the 5 V class LMBs unexplored.

On the other hand, traditional organic carbonate has a higher theoretical decomposition potential (6 V) and hence is anticipated to be an ideal platform for high-voltage LMB electrolytes.^{14,19,20} Other concomitant advantages of the carbonate electrolyte are also intriguing and include high thermal stability, superior conductivity, good compatibility with electrodes and high industrial maturity. Nonetheless, in reality, the decomposition potential of the carbonate electrolyte in high-voltage LMBs will decrease to around 4.3–4.5 V because of the catalytic reaction at the electrodes.²¹ The active dendrite growth on Li metal further deteriorates the coulombic efficiency and cycling stability.²² Prevailing strategies to tackle these challenges include electrolyte doping and electrode coating,²³ whereas the stability improvement is limited for 5 V LMBs (typically below 200 cycles). Herein, we report a molecularly passivated carbonate electrolyte for stable cycling of 5 V LMBs by introducing tris(pentafluorophenyl)borane (TPFPB) as the dual-effect additive. TPFPB is a boron additive that is composed of a boron central atom and a peripheral pentafluorobenzene group. The redox reaction of TPFPB can create boron-containing interfacial films on both the cathode and the anode. More critically, TPFPB has a higher HOMO energy and a lower LUMO energy compared with the conventional carbonate electrolyte solvents (*e.g.*, EC/EMC), leading to energetically favorable formation of artificial protective layers compared with the redox reaction of the standard electrolyte (STD). With the incorporation of TPFPB in EC/EMC electrolyte, we achieved simultaneous stabilizations of the 5 V LNMO cathode and the Li metal anode without sacrificing EC/EMC solvents.

Experimental section

Electrode preparation

LNMO powder was purchased from Wujie Science & Technology Co., Ltd., China. The cathode electrodes were made by casting a homogeneous slurry consisting of 80 wt% LNMO, 10 wt% Super P, and 10 wt% poly(vinylidene fluoride) (PVDF) on a flat aluminum foil with a glass bar; *N*-methyl pyrrolidinone (NMP) was used as the solvent. The mixture was uniformly mixed in an agate mortar and then, the electrode was dried at 60 °C for 2 h and vacuum-dried at 80 °C for 12 h. The cathode loading was $6 \pm 0.2 \text{ mg cm}^{-2}$. Before assembling, the electrodes were calendared at a pressure of 6 MPa.

Electrolyte preparation and cell assembly

The standard electrolyte (STD) was prepared by dissolving 1 M lithium hexafluoro phosphate (LiPF_6) in a solvent mixture containing 3 : 7 wt% ratio of ethylene carbonate and ethyl methyl carbonate (EC/EMC, 99.99% purity, Henan Huarui Co., Ltd., China). The additive TPFPB was added to the electrolyte in

amounts of 1 wt%. CR-2032-type coin cells were assembled with the LNMO composite electrodes in an Ar-filled glovebox, in which H_2O and O_2 concentrations were controlled to be less than 0.1 ppm. Li-metal foil was used as the counterpart and polypropylene (Xinxiang Zhongke Science and Technology Co., China) was used as the separator. Also, 50 μL electrolyte was used in each cell. The thickness and the diameter of Li-metal were 0.25 mm and 16 mm, respectively.

Characterizations and measurements

The charge–discharge performances of LNMO/Li half-cells with and without TPFPB were tested on a LAND cell-test system (Land CT 2001A, China) at room temperature. The cells were charged to 4.95 V and discharged to 3 V at 0.1C ($1\text{C} = 147 \text{ mA g}^{-1}$) for the three initial cycles, followed by cycling at 0.5C for 500 cycles. The cell voltage was stabilized after the third and 200th cycle according to the following protocol: charging to 4.95 V at 0.1C followed by a constant voltage at 4.95 V until the current decreased to 10% of the applied charging current. Then, electrochemical impedance spectroscopy (EIS) measurements were carried out at a Chenhua electrochemical station (Shanghai, China); the voltage amplitude was 10 mV and the frequency varied from 10^5 to 0.01 Hz. A three-electrode cell was assembled in an Ar-filled glove box with Pt as the working electrode and Li foils as the counter and reference electrodes, with which linear sweep voltammetry (LSV) measurements were performed at a Chenhua electrochemical station at a rate of 0.1 mV s^{-1} with an open-circuit voltage from 7 to 1.5 V. Rate performance measurements were carried out using the following protocol: 0.1C for the initial three cycles, then cycling at 0.5, 1, 3, 5, 8, and 0.1C separately. Chronoamperometric responses were measured on the aforementioned LAND cell-test system before the test cells were charged–discharged three times and then kept at 4.95 V for 20 d.

The cells were disassembled in an Ar-filled glove box after cycling. Electrode samples were harvested from the discharged half-cells, washed three times with anhydrous dimethyl carbonate (99.99% purity, Henan Huarui Co. Ltd., China) to eliminate sediments from the electrode surface, such as LiPF_6 , and then vacuum-dried for 8 h at room temperature before analysis. Surface morphologies of the fresh and cycled electrodes were characterized with FE-SEM (SUPRA40, Zeiss GmbH, Germany) and the microstructures of the fresh and cycled LNMO powders were characterized by JEM-2100 (JEOL Corp., Japan). The surfaces of the cathode and anode were analyzed using X-ray photoelectron spectroscopy (XPS) on an ESCA-LAB250Xi X-ray photoelectron spectrometer.

Theoretical calculations

The oxidation and reduction tendencies of organic molecules can be simulated by calculating the highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) energy levels. The HOMO/LUMO energies for solvents EC and EMC and the additive TPFPB were calculated based on density-functional theory (DFT) using the B3LYP functional and

6-311G++ (d,p) levels of the basis set in the Gaussian 09 framework.

Results and discussion

We first calculated the chemical structure and HOMO/LUMO energies for EC/EMC solvents and TFPFB additive using density-functional theory (Fig. 1a). The HOMO energy of TFPFB (-7.89 eV) is substantially higher than those of EC (-8.44 eV) and EMC (-8.16 eV) solvents, implying higher tendency to be oxidized at the LNMO cathode. The LUMO energy of TFPFB (-3.54 eV) is much lower than those of EC (0.54 eV) and EMC (0.82 eV),^{14,24} suggesting that TFPFB can be preferentially reduced on Li anode. Such an energy landscape is attractive since it guarantees the early redox reaction of TFPFB over EC/EMC, which facilitates the formation of a stable electrode/electrolyte interfacial layer on both electrodes and prevents the decomposition of EC/EMC.

Fig. 1b and c present the linear sweep voltammetry (LSV) results of EC/EMC electrolytes with TSTD and STD electrolytes using Pt as the working electrode and Li as the counter electrode at a scan rate of 0.5 mV s⁻¹. The oxidation current of STD (black line in Fig. 1b) starts from ~ 6.0 V and abruptly increases from ~ 6.5 V due to the intensive oxidation of carbonate

solvents.²⁵ As expected, for the TSTD electrolyte, there is a new oxidative peak at ~ 5.6 V, corresponding to the oxidation of TFPFB. This result confirms that TFPFB can be preferentially oxidized compared with the carbonate solvents. This phenomenon is further confirmed by the dQ/dV plot (Fig. S1†). In the case of STD, the peak located at ~ 4.05 V is associated with the oxidation of Mn^{3+} to Mn^{4+} .^{26,27} In the case of TSTD, a strong peak is observed at ~ 4.1 V, corresponding to the oxidation of TFPFB. The LNMO electrode used in the LSV test has superior catalytic property for additive oxidation compared with the Pt electrode used in the dQ/dV measurement. Therefore, a lower decomposition voltage of TFPFB is obtained in the dQ/dV curve than that in LSV measurement.^{28,29} On the other hand, the reduction of STD electrolyte occurs at ~ 2.1 V, whereas the reduction of TFPFB starts earlier at ~ 3.1 V (Fig. 1c), suggesting that TFPFB is reduced at the anode prior to carbonate solvents. These redox potentials correlate well with the calculated HOMO/LUMO energies of TFPFB and EC/EMC, supporting the formation of stable protective layers for both electrodes without losing carbonate electrolyte.

The cycling performances and coulombic efficiencies of LNMO/Li batteries using electrolytes with the TSTD electrolyte and the STD electrolyte were evaluated and compared (Fig. 2). Before cycling at higher charge-discharge current densities,

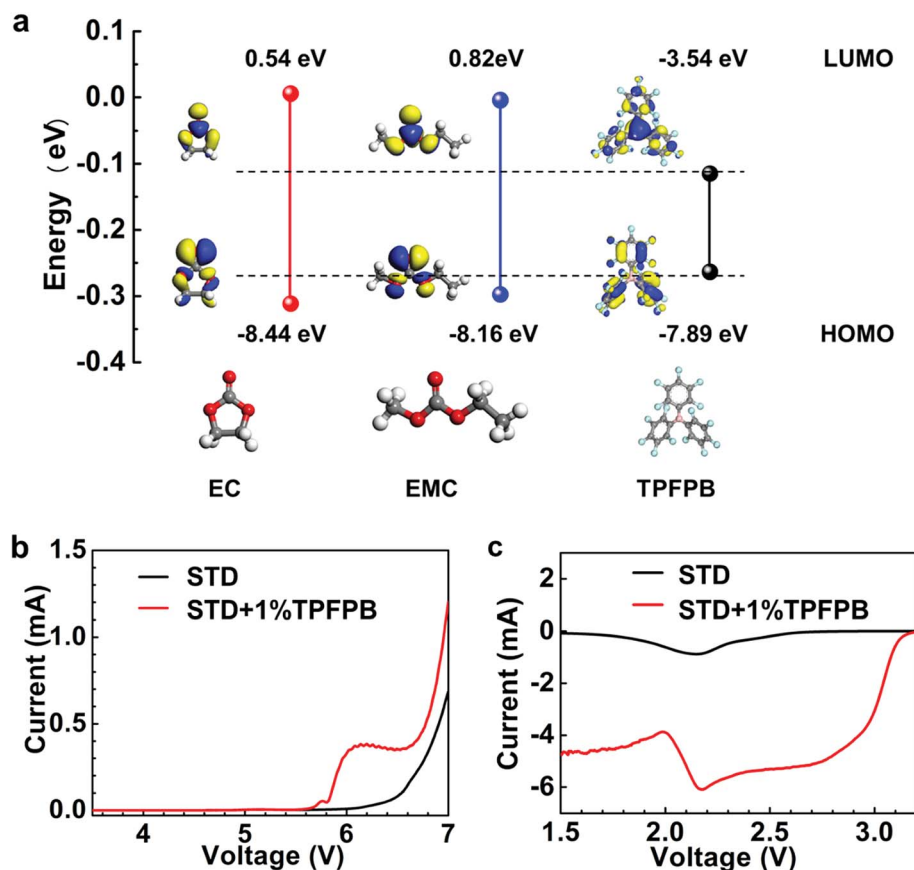


Fig. 1 Calculated energy landscape and redox potentials of EC, EMC and TFPFB. (a) Calculated HOMO and LUMO energy levels of EC, EMC, and TFPFB molecules. (b) Oxidation and (c) reduction linear sweep voltammetry of the EC/EMC electrolyte with TFPFB-containing (TSTD) electrolyte and STD electrolyte.

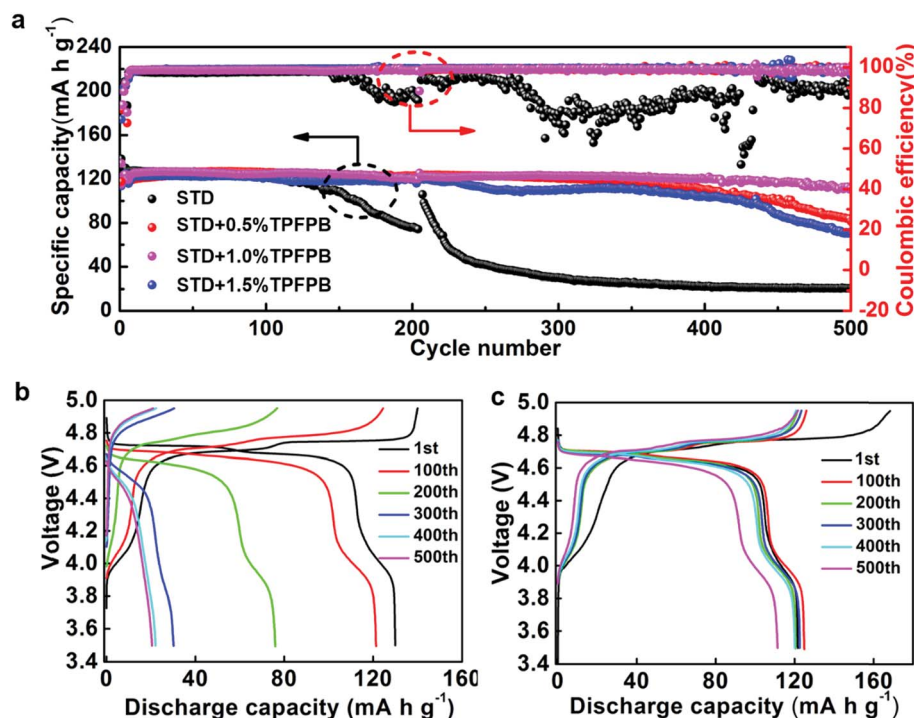


Fig. 2 Electrochemical performance of STD and TSTD. (a) Cycling performance of LNMO/Li batteries with STD and TSTD electrolytes. (b and c) Voltage profiles of LNMO/Li cells using (b) STD and (c) TSTD electrolytes.

three formation cycles were conducted at 0.1C ($1C = 147 \text{ mA g}^{-1}$) to convert TPFPB into protective layers. During the formation cycles, the cell is anticipated to exhibit lower coulombic efficiency for the TSTD electrolyte as substantial charges can be consumed by TPFPB conversions. Indeed, the coulombic efficiencies of 1.0% TSTD and STD electrolytes were 50.9% and 78.5%, respectively, with the initial charge-discharge capacities of 245.0 and 124.6 mA h g^{-1} and 162.5 and 127.2 mA h g^{-1} , respectively (Fig. 2a). After the formation cycles, the TSTD electrolyte could yield excellent cycling performance in an LNMO/Li half-cell cycled at 0.5C with a voltage window ranging from 3.5 to 4.95 V. After 500 cycles, the discharge capacity of STD decreases to 21.1 mA h g^{-1} , referring to a corresponding capacity retention of 17%. With the 1.0% TPFPB additive, the discharge capacity remains at a high level of 111.8 mA h g^{-1} , referring to a corresponding capacity retention of 90%, i.e., 5 times improvement. The cycling performances of both cells are interrupted at the 200th cycle due to impedance measurements, causing a fluctuation of specific capacity for the STD electrolyte. The coulombic efficiency of cells with the STD electrolyte fluctuates after 140 cycles; the coulombic efficiencies for 1.0% TSTD electrolyte remain at over 99% across the entire 500 cycles. The electrolytes with 0.5% and 1.5% TPFPB show improved performance compared with STD electrolyte, whereas inadequate or superfluous decomposition products cause stability deterioration compared with that observed for 1.0% TPFPB. The cycling stability of 1.0% TSTD electrolyte substantially outperforms the values for other commonly used additives such as diphenyl ether (DPE) and tri(hexafluoro-iso-propyl) phosphate (HFIP) (Fig. S2†).

Fig. 2b and c illustrate the voltage curves as a function of cycle numbers for LNMO/Li batteries with different electrolytes. Nearly no plateau exists in the voltage curves of the STD electrolyte after 300 cycles. In contrast, the plateau in the charge-discharge curves for the TSTD electrolyte remains steady after 500 cycles. In the dQ/dV curve of the STD electrolyte (Fig. S3a†), the oxidation peaks shift to higher voltages and the reduction peaks move to lower voltages upon cycling as a result of the ever-increasing polarization of the STD electrolyte. This result implies that most of the cathode active materials lose their efficacy after 300 cycles. The oxidation and reduction peaks remain constant for the cell cycled in the TSTD electrolyte after 500 cycles, further evidencing the remarkable cycling stability (Fig. S3b†).

To investigate the effect of TPFPB on individual electrodes, we decoupled the cathode and anode by harvesting LNMO and Li metal from cells after the initial three cycles in the TSTD electrolyte and pristine STD electrolyte (denoted as T_{cathode} , T_{anode} and S_{cathode} , S_{anode} , respectively). They were reassembled with new Li anodes and LNMO cathodes using STD as the electrolyte. Batteries built from T_{cathode} and T_{anode} electrodes show better cycling performances than those built with S_{cathode} and S_{anode} electrodes (Fig. S4†), indicating that TPFPB is beneficial for both the Li metal anode and the high-voltage LNMO cathode. The improvement in the cathode side is more substantial, suggesting that the dominant impact of TPFPB lies in the cathode side during the first three activation cycles. In general, the cycling performance of batteries with continuing TSTD electrolyte is better than those of the reassembled batteries with STD electrolyte probably due to the gradual

passivation of Li anode during the long-cycle operation. These results verify that the good cycling performance of LNMO/Li cells is because of the synergistic passivation of LNMO cathode and Li anode. TPFPB can also upgrade the rate capability of the high-voltage Li metal battery. The LNMO/Li cells using TSTD electrolyte also show a better discharge capacity compared to those containing STD electrolyte (Fig. S5†).

It is well documented that the traditional organic carbonate electrolytes usually result in resistive and unstable interfacial films on a high voltage cathode. LiF, Li₂CO₃, and other organic insulating by-products tend to accumulate on the electrode surfaces. To study the interfacial characteristics of such electrodes, electrochemical impedance spectroscopy (EIS) was carried out with the TSTD and STD electrolytes after 3 and 200 cycles. As shown in Fig. S6,† each curve contains two semicircles and one slope from high to low frequency, corresponding to the surface film resistance (R_{film}),³⁰ charge-transfer resistance of the electrode/electrolyte interface (R_{CT}),³¹ and the “Warburg” resistance.³² After the 1st and 3rd cycle, the impedance values for STD and TSTD electrolytes remain at a similar level (Fig. S6a and b†). However, R_{film} of batteries is considerably higher with the TSTD electrolyte after the 5th cycle (Fig. S6c†), which is a result of the formation of the protective film in TSTD electrolyte. After 200 cycles, the R_{film} and R_{CT} values of cells in STD electrolyte show distinct increase. However, this unwanted increment is reduced effectively after the addition of TPFPB (Fig. S6d†). This implies the successful reduction of resistive components when TPFPB is present.

We further conducted a series of material characterizations to disclose the origin of TPFPB benefits. Starting with the cathode side, Fig. 3 presents the field-emission scanning electron microscopy (FE-SEM) images for the fresh and cycled LNMO particles. There are two kinds of morphologies in the fresh electrode: sphere secondary particles and octahedral crystal shape (Fig. 3a and b). In STD electrolyte, the particles fragmented and pulverized after 200 cycles (Fig. 3e, f and S7†). However, both kinds of particles are well protected after 200 cycles in the TSTD electrolyte; also, no cracks or broken grains are observed (Fig. 3i, j and S8†). Trace amounts of moisture and acid in the electrolyte are unavoidable during the cell assembly. For the STD electrolyte (Fig. 3g and h), the corrosion by acid and moisture to the boundaries of crystals leads to the pulverization and cracking of LNMO particles.^{28,33} In contrast, the smooth surface of the protective film constructed by TPFPB (Fig. 3k and i) effectively separated LNMO from the electrolyte, acid and moisture. Cracking and pulverization are therefore prevented.

The TEM characterizations further identify the cathode electrolyte interface (CEI) layers for LNMO cathodes cycled with different electrolytes. There is no thin film encapsulation on the fresh LNMO particles (Fig. 3c and d). The LNMO electrode cycled in STD electrolyte forms thick and non-uniform CEI. The thickness of this CEI layer ranges from 59 to 178 nm, which is a consequence of the large amount of electrolyte decomposition on the electrode surface. The electrical isolation can even fully deactivate some LNMO particles during the charge–discharge reaction.³⁴ Consequently, delithiation and lithiation of LNMO are impeded, resulting in increase in impedance and

deterioration of cycling performance. The detrimental effect of thick and nonuniform CEI may be further amplified by particle fissure (Fig. 3g and h). This fissure can deteriorate the contact between particles and hinder the transport of Li ions and electrons, leading to increase in R_{CT} . On the other hand, the LNMO cathode cycled in TSTD electrolyte is covered with a uniform and thin layer with film thickness of approximately 10 nm (Fig. 3k and l). The high uniformity ensures thorough isolation of the LNMO particle from the carbonate electrolyte, leading to recovery of high intrinsic decomposition potential of carbonates and thus the prevention of electrolyte decomposition. In the meantime, low film thickness allows effective transport of Li ions and avoids the increase in cell impedance. The continuing decomposition of carbonate electrolyte in the STD sample can be visualized by the residual current in chronoamperometric response tests (Fig. S9†). The residual current caused by the electrolyte decomposition is about 0.006 mA for STD electrolyte, and it reduces to nearly zero in TSTD electrolyte.

TPFPB also dramatically changed the surface chemistry of LNMO cathodes. Fig. 3m–o show the X-ray photoelectron spectroscopy (XPS) spectra of O, F and B on different LNMO particles. For pristine LNMO, the O 1s peak centered at ~529.6 eV belongs to the M–O (M = Mn, Ni) bond (Fig. 3m);²⁹ it disappears in the spectra for both STD and TSTD electrolytes owing to the formation of CEI layers, which covered the M–O bonds. The peak located at ~534 eV is the signal of poly(CO₃) created by the ring-opening polymerization of EC in STD electrolyte under electrochemical action, which is also the indication of electrolyte decomposition.²² However, this signal is not detected for the TSTD electrolyte. These results confirm that the decomposition of solvents in TSTD electrolyte is effectively suppressed by the TPFPB additive. In the F 1s spectra, the LNMO cathode exhibits two main peaks at ~687.8 and ~685 eV, corresponding to the C–F bond in poly(vinylidene fluoride) (PVDF) and the Li–F bond. The peak intensity for the LiF bond is moderately weaker in the TSTD electrolyte compared with that in the STD electrolyte. This is caused by the accelerated dissolution of LiF when TPFPB is present as a result of the interaction between B (anion receptor) and F.^{35,36} Since LiF is the main resistive substance that hinders Li transport during cycling, reducing LiF accumulation on the surface of TSTD electrode will substantially enhance the transport of Li ions, as detected by the EIS test (Fig. S6†). In the B 1s spectra, an apparent peak at ~192 eV can be observed in the TSTD electrolyte but not in the STD one, verifying that TPFPB has participated in the formation of CEI. These results justify that TPFPB can effectively relieve the oxidation of electrolyte at the cathode side.

In addition to cathode protection, TPFPB also shows outstanding passivation performance on the Li anode side. Fig. 4a–f demonstrate the top and cross-sectional view images of pristine, STD and TSTD Li anodes after three reaction cycles. Fresh Li exhibits a smooth and clean surface (Fig. 4a and b). For the STD case, the top view SEM image reveals dendritic and mossy Li growth on the electrode surface (Fig. 4c). In addition, a porous and rough surface with a thickness of 38 μm is observed in the cross-sectional view (Fig. 4d), indicating that

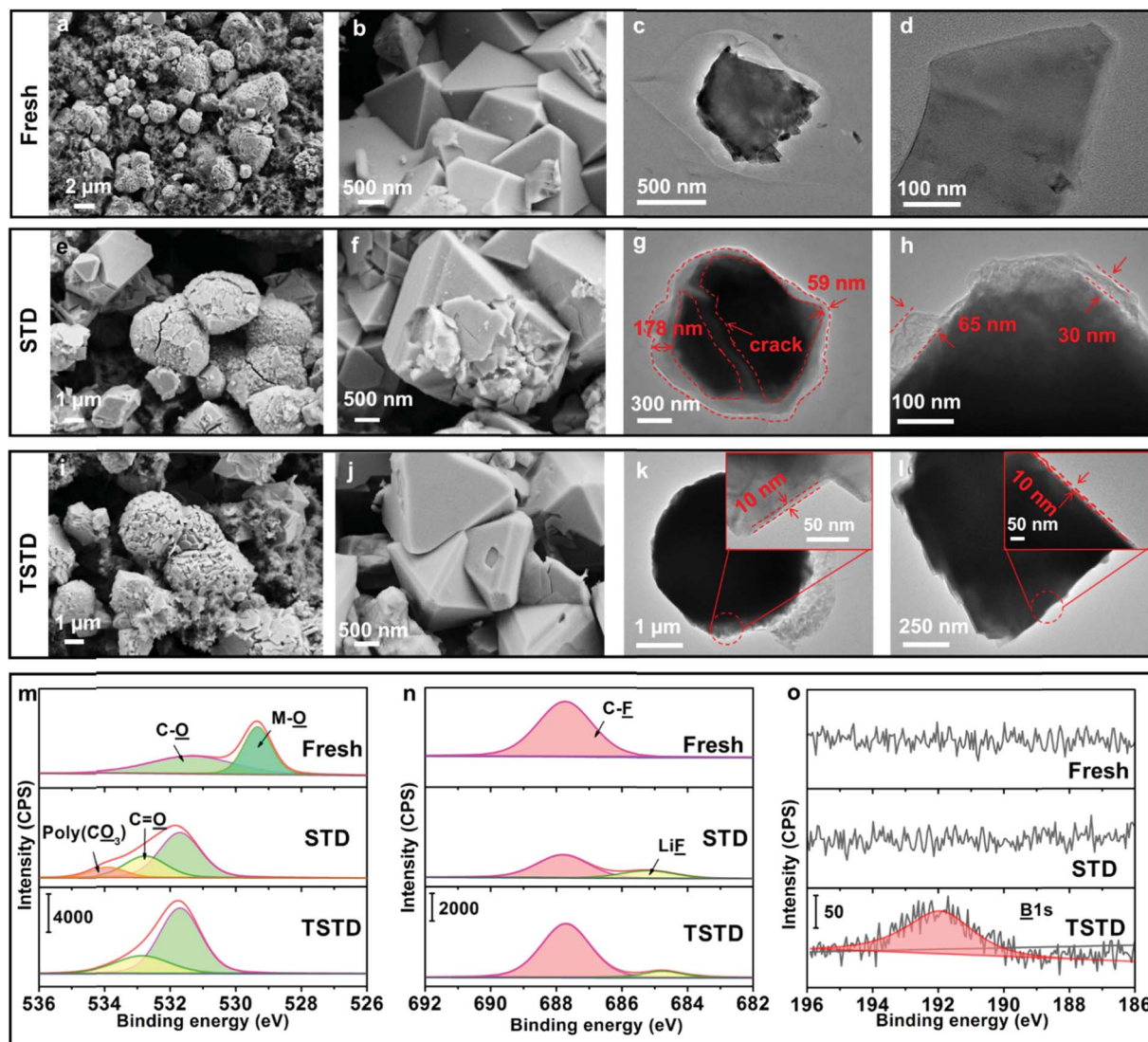


Fig. 3 Electron microscopy and elemental characterizations of the LNMO cathode. (a–l) SEM (left) and TEM (right) images of the LNMO electrodes under the conditions of (a–d) pristine LNMO cathode and after 200 cycles in (e–h) STD and (i–l) TSTD electrolytes. XPS spectra of (m) O 1s, (n) F 1s, and (o) B 1s for LNMO cathodes, fresh and after cycling in STD and TSTD electrolytes.

a significant amount of electrolyte has decomposed and a considerable amount of “dead Li” is formed. This can lead to substantial unwanted polarization and micro-short-circuits. These results indicate that the traditional carbonate electrolyte cannot form an effective solid-electrolyte-interface (SEI) layer, and Li cannot be deposited uniformly. In contrast, an evident SEI was observed on the Li anode when TPFPB was added to the electrolyte (Fig. 4e). The Li anode has a dense and smooth surface without any dendrites (Fig. 4f), indicating that negligible carbonate experienced solvent decomposition during the process of Li deposition. To thoroughly study the long-term influence of TPFPB, SEM images of the Li anode after 200 cycles were recorded and compared (Fig. S10†). The top view SEM image displays a large amount of dendrites for the STD electrolyte. For the TSTD electrolyte, a smooth, compact, and dendrite-free surface is observed. Above observations confirm the beneficial role of TPFPB in promoting the

uniform deposition of Li and reducing the formation of dendrites.

Similar to the cathode, the Li anode surfaces of STD and TSTD samples terminate in distinctive chemical compositions. Fig. 4g–i show the high-resolution C 1s, F 1s, and B 1s XPS spectra of the anodes after the initial three cycles. For Li anode cycled with STD electrolyte, the surface components of C 1s spectra include the peaks for C–H/C–C located at ~284.6 eV, C–O at ~285.2 eV, C=O at ~286.8 eV, and poly(CO₃) at ~289.5 eV (Fig. 4g).^{22,37} The poly(CO₃) constituent represents the typical components derived from the reduction of carbonate-based electrolyte.³⁸ For the TSTD sample, the poly(CO₃) signal at ~289.5 eV is substantially decreased, indicating the suppression of electrolyte decomposition. In the meantime, F 1s spectra exhibit higher Li_xPO_yF_z and C–F signals for STD electrolyte because of the reaction between LiPF₆ and moisture, which produces Li_xPO_yF_z and HF. HF can then attack the

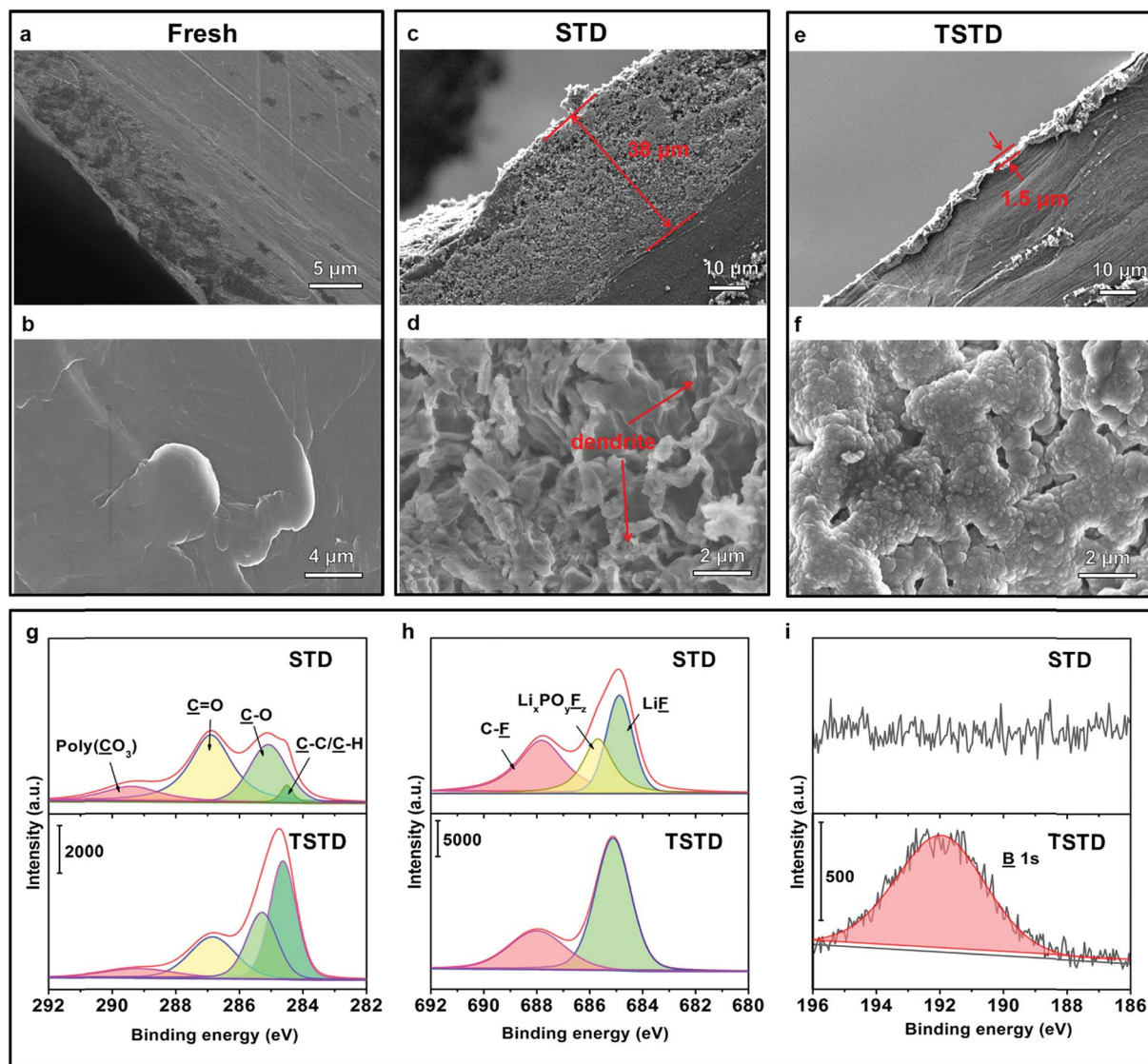


Fig. 4 Electron microscopy and elemental characterizations of the Li anode. (a–f) Top and cross-sectional view SEM images of the Li anodes under the conditions of (a and b) fresh surface and after initial three cycles in (c and d) STD and (e and f) TSTD electrolytes. (g–i) XPS spectra of (g) C 1s, (h) F 1s, and (i) B 1s for fresh Li anodes and after being cycled three times in different electrolytes.

carbonic solvents, leading to serious decomposition of the STD electrolyte. Conversely, $\text{Li}_x\text{PO}_y\text{F}_z$ peak mostly disappears for TSTD electrolyte and the C–F peak intensity is much weaker. Unlike the observations for cathode, the LiF content on the anode cycled with TSTD electrolyte is slightly higher than that cycled with STD. According to Fig. 1, the discrepancy in the LUMO levels between TPFPB and EC or EMC is far more substantial than that in the HOMO levels. Such a HOMO and LUMO arrangement indicates that TPFPB is more productively reduced at the anode compared with its oxidation at the cathode. For this reason, the protective film on the anode (Fig. 4e) is much thicker than that on the cathode (Fig. 3k and l); with the use of XPS, we detected more B signals on the anode than on the cathode (Fig. 3o and 4i). The TPFPB compound possesses large fluorine content in the

pentafluorophenyl moiety. After its preferential reduction at the anode surface, the fluorine was released and bonded with Li^+ , leading to LiF accumulation on the Li metal surface compared with the observations in the case of STD. In general, insulating LiF is soluble in TPFPB. However, we introduced a fairly small amount (1%) of TPFPB to the electrolyte. After formation cycles, most of the TPFPB molecules were consumed to construct the solid protecting film on the electrode surfaces. The dissolution caused by the few remaining TPFPB species in the electrolyte is insufficient to influence the thick protective film formed on the anode. On the other hand, the remaining TPFPB species might be able to dissolve a sizable portion of the thin protective layer formed on the cathode and therefore slightly reduce its LiF content compared with that in the case of STD.

Conclusion

In summary, we developed a stable carbonate electrolyte system for 5 V Li metal batteries by introducing TFPFB as the additive. DFT calculations show that the TFPFB molecules possess a higher HOMO level and a lower LUMO level compared with EC/EMC. Such an energy landscape facilitates preferential oxidation of TFPFB on LNMO cathodes and early reduction on Li anodes. These reactions produced thin and uniform protective layers on both the cathode and the anode. The artificial interfaces built on the cathode surface reduced the side reactions between the cathode and the electrolyte and thus prevented the decomposition of carbonate electrolyte and avoided the pulverization of the LNMO cathode. The protective layer on the anode surface suppressed Li dendrite growth. The small film thickness minimized the system impedance. The 1% mass addition of TFPFB improved capacity retention of LNMO/Li batteries from 17% to 90.2% after 500 cycles with voltage cut-off of 4.95 V, average coulombic efficiency of 99% and very limited polarization. Beyond the high-voltage Li metal battery, this passivated carbonate electrolyte is potentially applicable to other high-voltage Li-metal battery systems.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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